The Physical Plane

A five-coordinate (x, y, z, t, S) mapping of morfs, selectors and replicators

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Abstract

The physical plane is mapped as a five-dimensional vector space (x, y, z, t, S), in which the fifth coordinate S is scale, the renormalisation scale μ taken as a real direction. Every physical structure (morf) is a vector in this space, and the vacuum it exists in (Darkness) is its counterpart. On the map, morfs occupy definite places along S from hadrons to galaxy clusters. The constraints that decide which morfs can persist (the Physical Selector) appear as boundaries, and self-copying structures (physical replicators) fall inside a window those boundaries define.

The rest of the analysis follows from this map. Given a geometry along S, the vacuum ("Darkness") becomes a medium whose curvature and energy change with position on the axis. Vacuum energy measured at the Planck, electroweak, QCD and cosmic positions falls along S in that order. The gravitating part scales as $M_p^2\mu^2$, so between the Planck floor and the cosmic slice it falls by $(M_p/H_0)^2 \approx 10^{120}$: the quantum-field and cosmological values belong to different points of one axis, and adding or comparing them directly is a category error.

In the galactic region of the map, the gradient between the vacuum relaxed by a galaxy and the unrelaxed vacuum around it acts as unseen mass. Written as a field equation, this gradient reproduces flat rotation curves and the baryonic Tully–Fisher relation across 153 galaxies, with a strength set by the cosmic vacuum. Tested on 145 galaxies at $z \approx 0.7$–2.5, that strength tracks the vacuum density rather than the expansion rate: the galactic scale is consistent with staying nearly constant and does not grow as $cH(z)/2\pi$ (excluded at 3.4σ). At the top of the map the vacuum is fading today, at 2.8σ in current cosmological data. In groups and clusters, the galaxies' own Darkness and the hot plasma's unrelaxed vacuum reproduce the observed masses. General relativity and the Standard Model are recovered wherever they are tested.

1 The physical plane as a vector space

Every physical structure has a place in space, a characteristic time and a characteristic size. The first two are coordinates in ordinary physics. Size is usually treated as a property. Here it is treated as a coordinate S of the same standing, which is the renormalisation scale taken as a real direction:

$$S = \frac{1}{k}\ln\frac{\mu}{\mu_0}, \qquad \text{structure} = (x, y, z, t, S)$$

In practice, each structure's S is the number of e-folds of its size above the Planck floor, and its t is its own dynamical (free-fall) time.

The Law of the Dark. The organising principle of the framework:

The vacuum medium (Darkness) carries negative curvature at every scale layer. This curvature becomes more negative at smaller S and flattens toward zero at larger S. The presence of matter (positive curvature) locally relaxes the surrounding negative curvature. The gradients created by this scale-dependent relaxation produce every force, every coupling hierarchy, and every cosmological phenomenon we observe.

The sections below follow it along the map. The geometry shows the relaxation producing gravity's strength (Section 3), the galactic dark-matter effect (Section 5) and dark energy (Section 6). The strong, weak and electromagnetic forces are the same Material Field at their own positions on S: moving along S is exactly renormalisation-group running (Section 3), so their measured running is the Material Field read along the axis.

Terms.

Term	Meaning
Darkness	The physical medium that makes up the vacuum at any scale layer. It is not an absence but a substrate of structures from lower S, and it carries negative curvature. Virtual particles are fluctuations of Darkness. Darkness is the counterpart of morfs: the medium in which structure exists.
Void	Localised negative curvature of the vacuum at a given place and scale; the geometric opposite of a mass concentration. Mass relaxes the surrounding curvature inward (attraction); a Void steepens it outward (repulsion). Every region is a superposition of mass and Void contributions at each S.
Morf	Any physical structure the Physical Selector allows at its scale layer, from standing waves and crystals to RNA molecules and nervous systems.
Physical Selector	The physics at a given S (curvature constraints, energy budgets, thermodynamic bounds) that determines which morfs can exist there. It is not a separate mechanism; it is the physics at that S.
Replicator	A unit of evolution that is copied with variation (see the taxonomy below).
Material Field	Gravity and the three Standard Model forces as one geometric field, projecting differently at each S: gravity at cosmic S, the dark-matter gradient at galactic S, and the electroweak, electromagnetic and strong forces at their own S.
Ethereal Field	The counterpart of the Material Field: the field of Darkness and its repulsive projections at each scale layer. At cosmic S it is dark energy; at QCD S it is the QCD vacuum energy.
Universal Field	The Material and Ethereal fields within the observable range of S: all of known physics.
Omniversal Field	The same two fields extended across all values of S, including those beyond the observable range.
Omniverse	The full extent of the Omniversal Field across all S. Our universe is one S-window within it. Because S is a coordinate with no privileged end, the axis continues beyond the observable floor and ceiling.
Scale layer	A value or narrow range of S at which physics takes a particular projected form.
S-dark energies	The Ethereal projections at different S values; values of one function at different places, not to be added together.
S-floor / S-ceiling	The lowest and highest S of the observable range. Neither is a physical infinity; crossing either is a transition into a different S-range.
Sign flip	Each tier's structures become the medium (Darkness) of the next tier: the ground of the next layer is the complexity of the previous one.
Law of the Dark	The organising principle: Darkness's negative curvature steepens at small S and flattens at large S, matter relaxes it locally, and the gradients of that relaxation produce the forces and the cosmological phenomena (Section 1).

Replicator taxonomy. Replicators are classified along three independent axes. On the physical plane the Type is always Morf.

Axis	Encodes	Values	On the physical plane
Type	what the replicator is made of	Morf (physical structure), Info, Qualia	Morf
Class	the domain of replication	Gene (natural), Meme (cultural), Cog (cognitive)	Gene; Meme-Morf for physical artifacts
Mode	how it persists	Exec (a running process that spreads by starting elsewhere), Perma (a fixed structure that spreads by being copied)	both

The approach is to build the map first and then ask what follows from it. Section 2 places structures on the map. Sections 3–6 show what the geometry along S implies for the vacuum, galaxies and the cosmos. Section 7 checks that known physics is recovered.

2 The map: morfs, the Selector and replicators along S

Placed on the S axis, the structures of the physical plane fall into tiers, bounded by lines the couplings predict, and the replicators occupy a window between those lines. S runs from 0 at the Planck floor to 139.9 e-folds at the Hubble ceiling.

Structure	S (e-folds)	Position on the axis	Density log ρ (kg/m³)
Proton	45.1	32%	17.8
Uranium nucleus	47.2	34%	17.4
Hydrogen atom	56.1	40%	3.4
Self-replicating RNA	60.6	43%	2.3
Virus	63.0	45%	3.3
E. coli	65.9	47%	2.4
Human	79.7	57%	1.4
Giant sequoia	83.6	60%	0.7
Neutron star	89.2	64%	17.6
Earth	95.4	68%	3.7
Jupiter	97.8	70%	3.1
Sun	100.1	72%	3.1
Globular cluster	120.0	86%	−16.8
Milky Way	128.5	92%	−21.7
Coma cluster	131.5	94%	−22.8

The Physical Selector. Two lines bound the map. The quantum line, $R = \hbar/Mc$, sets how small a mass can be confined. The black-hole line, $R = 2GM/c^2$, sets how compact it can be. They meet at the Planck floor. All 22 catalogued structures lie inside the triangle they enclose.

Tiers from the couplings. Running couplings place each tier boundary where it is observed:

Boundary on the map	Predicted	Observed	Offset (dex)
Hadron size, where $\alpha_s = 1$	6.9×10^{-16} m	8.4×10^{-16} m	0.09
Nuclear density line	1.2×10^{18} kg/m³	3.9×10^{17} kg/m³	0.49
Condensed-matter density line	2.7×10^{3} kg/m³	4.3×10^{3} kg/m³	0.20
Largest cold body, $\sqrt{(\alpha/\alpha_G)} \cdot a_0$	5.9×10^{7} m	7.0×10^{7} m	0.07
Stellar mass, $\alpha_G^{-3/2}\, m_p$	3.7×10^{30} kg	2.0×10^{30} kg	0.27

Morfs cluster on two density lines, nuclear (log ρ = 17.6) and condensed (log ρ = 2.4). Each line is set by the force that dominates that stretch of S: the strong force below the hadron

boundary, electromagnetism above it. Gravity overtakes electromagnetism at the planet–star boundary.

Replicators. Replication needs chemistry, which requires sizes above about 10 a_0, and it needs freedom from gravitational crushing, which requires sizes below $\sqrt{(\alpha/\alpha_G)}\cdot a_0$. That window spans S = 58–98. Every catalogued replicator, from self-replicating RNA to the largest organisms, lies at S = 61–84, inside it.

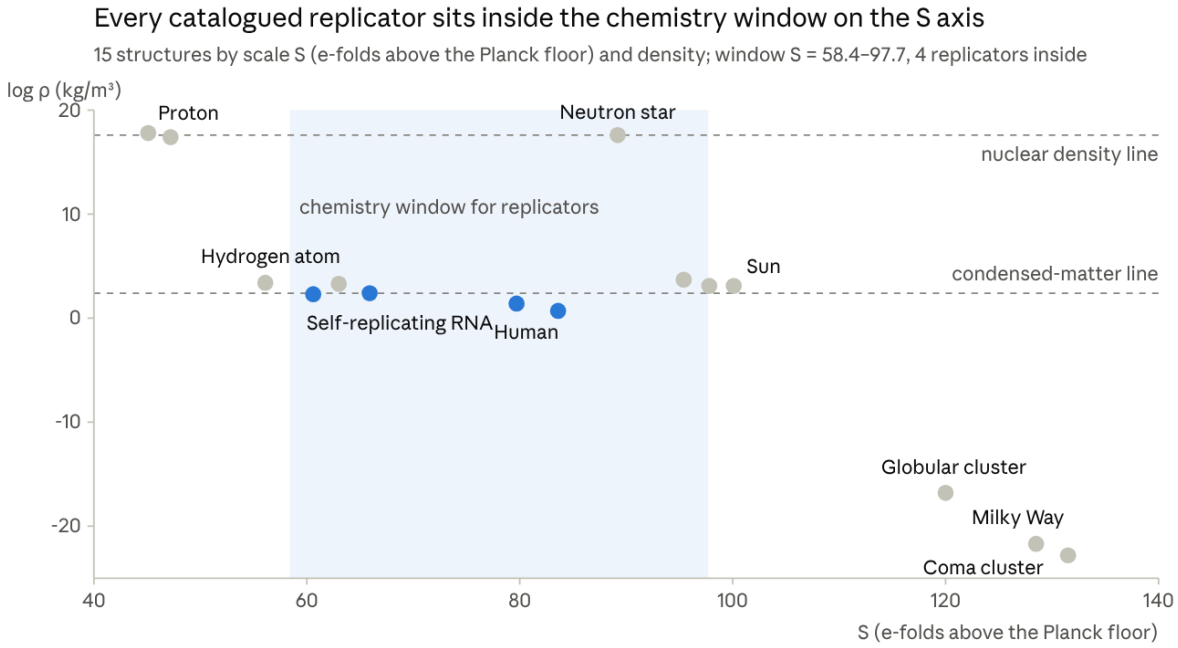


Figure 1 · catalogue of Section 2 (test suite, cell p03) · window from 10 a_0 to $\sqrt{(\alpha/\alpha_G)}\cdot a_0$

Morphological evolution. The tiers follow the Law of the Dark at successively higher S: hadrons to nuclei, nuclei to atoms, atoms to molecules, molecules to self-replicating structures. Each tier's stable structures become the Darkness of the next. With a Physical Selector at each S, selection, variation and replication give the successive stabilisation of more complex structures as S increases: morphological evolution read as a consequence of the geometry rather than an added assumption.

The tier boundaries agree with standard scaling arguments (Carr & Rees 1979; Press & Lightman 1983). What the map adds is a single coordinate on which all of them, and the replicators, can be read together.

3 The geometry behind the map

If S is a coordinate, the map needs a metric. The simplest metric with S as a direction is warped, of the Randall–Sundrum type (Randall & Sundrum 1999). Each identity below was verified symbolically.

$$ds^2 = e^{2A(S)}\left(-dt^2 + a(t)^2 d\vec{x}^{\,2}\right) + dS^2$$

- **Darkness is negatively curved.** Flat slices of fixed S require a negative bulk vacuum, $\Lambda_5 < 0$. This is anti-de Sitter.
- **The vacuum seen at a given S** is $\Lambda_{\text{eff}}(S) = \Lambda_5/2 + 3A'(S)^2$. It has a negative bulk part

and a positive part from how the geometry bends with S, so it changes with position on the map.

- **Matter relaxes Darkness.** Energy on a slice raises A'^2 and offsets the negative bulk term.
- **Sign flip.** Across a slice that carries energy, the bending reverses sign, and the slice's own vacuum term changes sign at a critical energy content.
- **Floor and ceiling.** The S-floor is where the quantum and black-hole lines meet, at $R = \sqrt{2}\,\ell_P$. The ceiling is the Hubble scale, where the universe lies on its own black-hole line. Neither edge ends the axis: S continues beyond both, so our universe is one S-window of the Omniverse. Read this way, the Big Bang is a crossing of the S-floor into our window, heat death is the approach to the S-ceiling, where our physics becomes the Darkness of a larger system, and black holes are floor crossings that can seed new S-windows.
- **S can move in time.** A homogeneous S(t) obeys consistent coupled equations with the expansion. A ghost-free S(t) cannot make the vacuum grow.

The same geometry fixes four more things:

- **Gravity's strength.** The junction condition at a slice that carries energy gives the slice's own Friedmann equation and its Newton constant, $G_4 = \kappa^4\sigma/48\pi$ (Binétruy, Deffayet & Langlois 2000). How strongly matter gravitates on a slice is set by the curvature of Darkness and the slice's energy content.
- **Moving along S is renormalisation-group running.** In the warped bulk $S = (1/k)\ln(\mu/\mu_0)$, so $d/dS = k\cdot d/d\ln\mu$, exactly the RG derivative. Couplings placed along S run as the Standard Model says, with no added term.
- **Gravitational waves travel at c.** A tensor perturbation's zero mode on a slice obeys $\Box h = 0$, so its speed is exactly c.
- **The tuning moves into the geometry.** A slice's own vacuum is the difference between the bulk term and the slice's energy content, and our small positive vacuum sits just above the flip. The cosmological-constant cancellation is carried by the geometry, not removed. In a literal fifth dimension it is one part in about 10^{62}; in the holographic reading the slice tension has no measured value.

The geometry is read holographically, as in the correspondence between a warped bulk coordinate and energy scale (Maldacena 1998). S is scale, not an extra direction of space, so space keeps its three dimensions.

4 The vacuum along the map

As the Law of the Dark requires, vacuum energy measured at four positions on the map falls from the bottom to the top of the axis: steeply at small S and gently at large S. The vacuum along S has two parts, and the table holds both.

Position on the map	**Scale μ**	**Vacuum energy (eV^4)**
Planck	2.4×10^{27} eV	3.5×10^{109}
Electroweak (Higgs potential)	2.5×10^{11} eV	1.2×10^{44}
QCD (gluon condensate)	2.4×10^{8} eV	3.4×10^{33}
Cosmic (observed Λ)	1.4×10^{-33} eV	2.4×10^{-11}

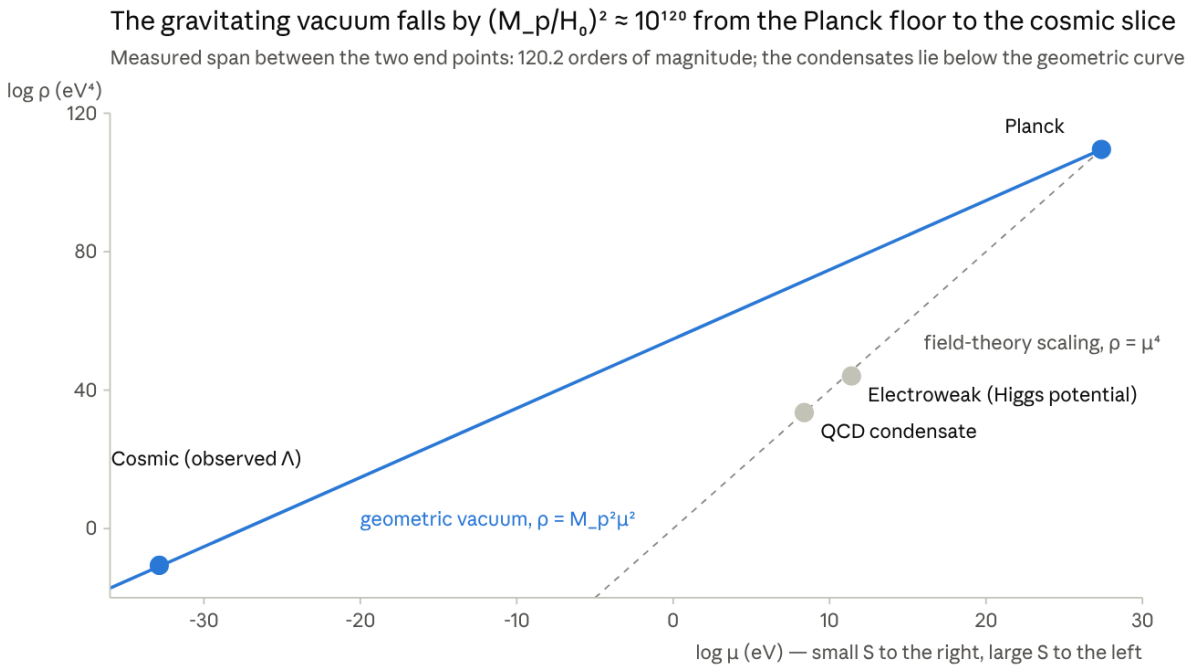


Figure 2 · vacuum values of the table above; curves from Section 3 (test suite, cells p02 and p05)

The gravitating vacuum. On each slice, the vacuum that gravitates is the geometric one of Section 3. Along the exact solution its curvature scales as the square of the slice's energy scale, $\Lambda_{\text{eff}} \propto \mu^2$, so its energy density is

$$\rho_{\text{vac}}(S) \sim M_p^2\, \mu^2(S) \qquad (\text{not } \mu^4)$$

At the Planck floor ($\mu = M_p$) this is M_p^4, the quantum-field value. At the cosmic slice, where $\mu = H_0$, it is $M_p^2 H_0^2$, the observed dark energy, within 0.3 dex. Between the two the vacuum falls by

$$\frac{\rho_{\text{vac}}(\text{floor})}{\rho_{\text{vac}}(\text{cosmic})} = \left(\frac{M_p}{H_0}\right)^2 = 2.9 \times 10^{120}$$

so the size of the discrepancy follows from the geometry. Placing our slice at $\mu = H_0$ is an identification of where our layer sits on the axis, not a derivation. The same μ^2 scaling appears in holographic bounds on the vacuum (Cohen, Kaplan & Nelson 1999); here it follows from the geometry of S, and the same vacuum sets the galactic scale of Section 5.

The condensates. The QCD and electroweak values are not the gravitating vacuum of those slices. They are the Darkness of their layers, made of the structures around them, and by the sign flip they are carried by those structures: most of a proton's mass is QCD energy. They gravitate as matter, not as vacuum, and they lie far below $M_p^2\mu^2$, as they must.

The cosmological constant problem as a category error. The quantum-field estimates and the observed cosmological constant are values at different places on S. Adding or comparing them directly is a category error, like adding the temperature of the Sun to the distance between two cities. Read on the map, their difference is the vacuum's fall along the axis, $(M_p/H_0)^2$, not a failed prediction. The fine tuning does not vanish: it is carried by the geometry (Section 3).

Sign flip on the map. If each position's Darkness is made of the structures just below it, the two should have comparable energy densities:

- at the QCD position, vacuum energy and nuclear-matter density agree within 0.30 dex;
- at the cosmic position, dark energy and matter agree within 0.36 dex today.

The cosmic match holds near the present epoch: the ratio of dark energy to matter was 0.29 at z = 1 and 0.002 at z = 9. Why the cosmic layer sits at its balance now is the coincidence problem, which the map restates rather than solves.

5 The galactic region of the map: the vacuum gradient

Around a galaxy the vacuum is relaxed where the galaxy's matter dominates and unrelaxed farther out. The gradient between the two acts as unseen mass.

Field equation. Let r(y) be the relaxed fraction of the vacuum where the matter's own pull is $g_N = y \cdot g\dagger$. The unrelaxed part acts through its gradient:

$$\nabla^2\Phi = 4\pi G\rho + \nabla\cdot\left[\left(\frac{1}{r(y)} - 1\right)\nabla\Phi_N\right], \qquad r(y) = 1 - e^{-\sqrt{y}}$$

The unseen mass density is the divergence of the unrelaxed vacuum. The equation has the quasi-linear form of Milgrom (2010), and the relaxation profile, the simplest one with the two required limits, matches the observed radial acceleration relation (McGaugh, Lelli & Schombert 2016):

- **Where matter dominates,** the vacuum is fully relaxed and gravity is Newtonian.
- **Where the vacuum is unrelaxed,** the response becomes scale-invariant. This follows if a shift along S is a rescaling: in three dimensions the invariant form is $|\nabla S|^3$. It gives $g \to \sqrt{(g_N\, g\dagger)}$, flat rotation curves and $v^4 = GMg\dagger$.
- **Unseen mass grows outward,** from zero inside to $M_{ph} \to \sqrt{(M\, g\dagger)}\cdot r$ outside, following the unrelaxed vacuum.

Data. The equation was tested on 153 SPARC rotation curves (Lelli, McGaugh & Schombert 2016), marginalising each galaxy's stellar mass-to-light ratio, distance and inclination. With the scale fixed by the cosmic vacuum, $g\dagger = cH_0/2\pi$, the equation describes the sample as well as with the scale left free ($\Delta BIC = -6.8$). The free fit gives 1.06×10^{-10} m/s².

The scale. g† is set by the cosmic vacuum, the same vacuum that appears as dark energy at the top of the map. That the galactic acceleration scale lies close to cH_0 was noted when it was first measured (Milgrom 1983); here it follows from the scale being set by the cosmic vacuum. Measured values lie at 0.11–0.29 × cH_0, depending on sample and on the stellar mass-to-light treatment, and rotation curves alone cannot separate $cH_0/2\pi$ from a fixed scale of 1.2×10^{-10} m/s²: the preference flips with the assumed mass-to-light ratio.

Redshift test. "Set by the cosmic vacuum" has two readings. If the scale follows the expansion rate, it grows into the past:

$$g_\dagger(z) = \frac{c\,H(z)}{2\pi}$$

reaching 1.8 times today's value at z = 1 and 3 times at z = 2. If it follows the vacuum density, which barely changes over that time, it stays nearly constant. The statistic $R = \ln[g\dagger(z)/g\dagger_{local}] \,/\, \ln[H(z)/H_0]$ is 1 for the expansion reading and 0 for the vacuum reading. The local scale is measured the same way on mass-matched SPARC discs, and the rule was fixed before running: a reading is favoured when R lies within 2σ of its value and more than 2σ from the other.

Sample	Galaxies	Redshift	R	Favours
Price et al. 2021 (dark-matter fractions at R_e)	41	0.7–2.5	+0.67 ± 0.23	expansion
KROSS (rotation and baryons at 3 R_e)	42	0.8–1.0	−0.37 ± 0.65	vacuum
KMOS3D (same)	41	0.7–2.2	−0.87 ± 0.55	vacuum
KGES (same)	21	1.3–1.6	−1.68 ± 1.52	inconclusive
Combined	145	0.7–2.5	+0.33 ± 0.20	vacuum

Combined, the galactic scale does not grow as cH(z)/2π: the expansion reading is 3.4σ away and the vacuum reading 1.7σ. This fits the vacuum reading: like the vacuum density, the scale is consistent with staying nearly constant over the last ten billion years. The samples disagree with each other (χ^2 = 9.9 for 3 degrees of freedom), with Price et al. alone pointing to growth. High-redshift discs are turbulent and partly pressure-supported, and several catalogue galaxies carry more baryons than their dynamical mass allows, so these systematics are not excluded. Section 5.2 carries the same mechanism to groups and clusters of galaxies.

5.2 Groups and clusters

In a cluster, the same vacuum gradient acts on more than one layer of the map at once, and two parts of the framework that are silent in a single disc galaxy come into play.

The galaxies' own Darkness (sign flip). A cluster's galaxies are the structures of the layer below it. Each keeps its own Darkness, the profile it has in isolation, out to the radius where it stops standing out from the surrounding matter (the Physical Selector): $\rho_D(r) = \sqrt{(M g\dagger/G)}/(4\pi r^2) = \rho_{matter}$. Counted over the measured galaxy population (stellar mass function of Vulcani et al. 2012, stars distributed with the measured concentration of van der Burg et al. 2014, measured stellar fractions of Kravtsov et al. 2018 and Eckert et al. 2022), this Darkness supplies the outer parts of clusters with no adjustable quantity.

Hot plasma. At 10^7–10^8 K the gas of groups and clusters is fully ionised. Its atomic structure is gone, so at the galactic layer it is not a morf: by the sign flip it feeds Darkness instead of relaxing it. Its thermal pressure therefore keeps the vacuum unrelaxed:

$$y_{\rm eff} = \frac{y}{1 + P_{\rm th}/P_0}$$

where matter is structured (disc galaxies, cold groups) P_{th} = 0 and the galactic law of Section 5 is unchanged. The pressure scale is measured once, from the 12 X-COP clusters (Eckert et al. 2022): $P_0 = 1.1 \times 10^{-12}$ Pa.

Results. With P_0 fixed, the mechanism was applied to systems not used to measure it.

System	Data	Residual (dex)
X-COP clusters, 12 (hot-gas masses)	Eckert et al. 2022	−0.08 / +0.07 / +0.06 / −0.08 (core, 200–500, 500–1000, >1000 kpc)
Clusters, 10 (galaxy-kinematic masses)	Li et al. 2023	+0.06 / +0.14 / +0.07 / −0.05
Relaxed X-ray groups, 15, cores ($r < 0.2\ r_\Delta$)	Gastaldello et al. 2007	+0.19 without the plasma term → +0.01 with it
Central cluster galaxies, 50 (MaNGA)	Tian et al. 2024	offset above the galaxy relation: observed +0.47 / +0.29 / +0.07, predicted +0.45 / +0.32 / +0.19

Cluster masses are reproduced within 0.15 dex in every region of two independent datasets, built with different methods (X-ray hydrostatic and galaxy kinematics). A stricter 0.1 dex

criterion, set before running, is met everywhere except the 200–500 kpc band of the kinematic set (+0.14 dex), where the two datasets disagree with each other by up to 0.2 dex for the same cluster (A2029). Group cores are filled, and central cluster galaxies sit above the galaxy relation by the amount their hot surroundings predict, running high at the strongest pull. The galaxies' Darkness carries the outskirts; the plasma carries the cores. The pressure scale P_0 is measured, not derived.

6 The top of the map: the fading vacuum

At the top of the axis the geometry requires the vacuum to flatten toward zero, and a ghost-free S(t) cannot make it grow. Current data agree: dark energy is fading today.

Data and result	**Value**
Dark-energy equation of state today, w_0 (SN + BAO + CMB + chronometers)	-0.85; constant vacuum disfavoured at 2.8σ
S(t) moving in the potential the geometry fixes	fits as well as a constant vacuum, $w \approx -0.95$ today
Long-term behaviour of that S(t)	vacuum energy tends to zero

The data are Pantheon+ supernovae (Brout et al. 2022), DESI DR2 BAO, Planck 2018 distance priors (Chen, Huang & Wang 2019) and 17 cosmic-chronometer H(z) measurements (Moresco et al., as compiled by López-Corredoira & Vazdekis 2018), each with its published uncertainties.

The vacuum that sets the galactic scale in Section 5 and the vacuum fading here are the same vacuum at two positions on the map. Dark matter and dark energy are the Material and Ethereal projections of one field at two values of S.

Two limits apply. The fitted history behind $w_0 = -0.85$ has dark energy below $w = -1$ before $z \approx 0.4$, which a ghost-free S(t) cannot follow: the data agree with the fading today, while the framework's own S(t) fits them as well as a constant vacuum, not better. And S(t) must start at today's dark-energy density, so it moves the "why now" coincidence into its initial condition rather than removing it.

7 Null tests

Wherever general relativity and the Standard Model are already tested, the mapping recovers them.

Null test	**Result**
Gravitational-wave speed	exactly c: the graviton zero mode obeys $\Box h = 0$ (Section 3); GW170817 allows $-3 \times 10^{-15} \le \Delta c/c \le 7 \times 10^{-16}$ (Abbott et al. 2017)
Atomic clocks	no drift predicted, since atomic couplings sit at their own position on S; measured $(1.8 \pm 2.5) \times 10^{-19}$ per year (Filzinger et al. 2023), 0.7σ from zero
Running couplings	moving along S is renormalisation-group running (Section 3), so Standard-Model running holds along the axis
Short-distance gravity	the holographic reading leaves inverse-square gravity unchanged; a literal fifth dimension would be bounded below 39 μm (Lee et al. 2020)
Nucleosynthesis	vacuum running acts as at most $\Delta N_{\text{eff}} \approx 0.02$
Growth of structure	BOSS $f\sigma_8$ (Alam et al. 2017): $\chi^2 = 1.24$ vs 1.28 for ΛCDM
Densities	every measured density lies at least 78 orders of magnitude below the Planck floor

Null test	Result
Black holes	a Planck-density core sits inside the horizon of every black hole above 4 × 10^{-9} kg; exteriors are unchanged from general relativity

8 Conclusion

Mapping the physical plane as (x, y, z, t, S) places every physical structure, the limits on which can exist, and the window for replicators on a single axis. Taking the axis as geometry, a vacuum that changes along it follows. Measured at four positions, that vacuum falls in the order the map requires, and its gravitating part falls by $(M_p/H_0)^2$ from the Planck floor to the cosmic slice. In the galactic region its gradient reproduces the rotation of disc galaxies with a strength set by the cosmic vacuum; in groups and clusters the sign flip and the unrelaxed vacuum of hot plasma reproduce the observed masses; and at the top of the map it is fading. Known physics is recovered wherever it is tested, and on 145 high-redshift galaxies the galactic scale is consistent with the vacuum density, not the expansion rate.

Data and code

All 73 results come from an open test suite with an executed notebook, run end to end on public data, downloaded with their checksums recorded. Every result is labelled: a derivation, a test with a rule fixed before running, a measurement, a constraint, a diagnostic, or a stated prediction. High-redshift rows are included in the repository, each checked against its catalogue. Nothing the code computes is discarded. Two earlier formulations, a linear galaxy law and a single vacuum curve through all four vacuum values, are kept with their pre-set rules and outcomes and marked as superseded by Sections 5 and 4.

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